



Student Number

St. Catherine's School Waverley

August 2014

TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics

General Instructions

- Reading Time – 5 minutes
- Working Time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- Show all necessary working in Questions 11-16
- Task weighting – 45%

Total Marks – 100

Section I

Pages 3-5

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section
- Answer on the multiple choice answer sheet provided.

Section II

Pages 6-13

90 marks

- Attempt Questions 11-16
- Allow about 2 hours and 45 minutes for this section
- Answer each question in the booklet provided.

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Section I

Total marks - 10

Attempt Questions 1-10

All questions are of equal value.

Answer either *A, B, C* or *D* on the multiple choice answer sheet provided.

Q1. Factorise $x^2y - xy^2 - x + y$.

(A) $(xy - 1)(x + y)$

(B) $(xy - 1)(x - y)$

(C) $(xy + 1)(x + y)$

(D) $(xy + 1)(x - y)$

Q2. Find $\int (2x + 1)^5 dx$

(A) $\frac{(2x + 1)^6}{3} + C$

(B) $\frac{(2x + 1)^6}{6} + C$

(C) $\frac{(2x + 1)^6}{12} + C$

(D) $\frac{5(2x + 1)^4}{4} + C$

Q3. Find the limiting sum of the geometric series

$$1 + \frac{\sqrt{2}}{\sqrt{2} + 1} + \frac{2}{(\sqrt{2} + 1)^2} + \frac{2\sqrt{2}}{(\sqrt{2} + 1)^3} + \dots$$

(A) $\sqrt{2} + 1$

(B) $\sqrt{2} - 1$

(C) $-\sqrt{2} - 1$

(D) $\frac{1}{\sqrt{2} + 1}$

Q4. For what values of k does the quadratic equation $kx^2 + kx + 1 = 0$ have no real roots?

(A) $k < 0$ or $k < 4$

(B) $0 < k < 4$

(C) $k > 4$

(D) $k < 0$ or $k > 4$

Q5. Solve the equation $\sqrt{3} \tan x + 3 = 0$ for $0 \leq x \leq 2\pi$

(A) $x = \frac{5\pi}{6}, \frac{11\pi}{3}$

(B) $x = \frac{\pi}{3}, \frac{4\pi}{3}$

(C) $x = \frac{2\pi}{3}, \frac{5\pi}{3}$

(D) $x = \frac{\pi}{6}, \frac{7\pi}{6}$

Q6. The table below shows the values of a function $f(x)$ for five values of x .

x	2	2.5	3	3.5	4
$f(x)$	4	1	-2	3	8

What value is an estimate for $\int_2^4 f(x)dx$ using Simpson's Rule with five function values?

(A) 12

(B) 6

(C) 8

(D) 4

Q7. If α and β are the roots of $2x^2 + 3x - 6 = 0$, what is the value of $\frac{\alpha\beta}{\alpha + \beta}$?

(A) 2

(B) $\frac{1}{2}$

(C) $-\frac{1}{2}$

(D) -2

Q8.

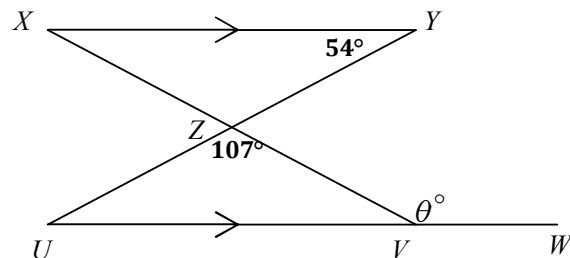


FIGURE NOT TO SCALE

The diagram above shows XY parallel to UW , $\angle XYU = 54^\circ$, $\angle UZV = 107^\circ$ and $\angle ZVW = \theta^\circ$. The value of θ is:

(A) 161

(B) 19

(C) 54

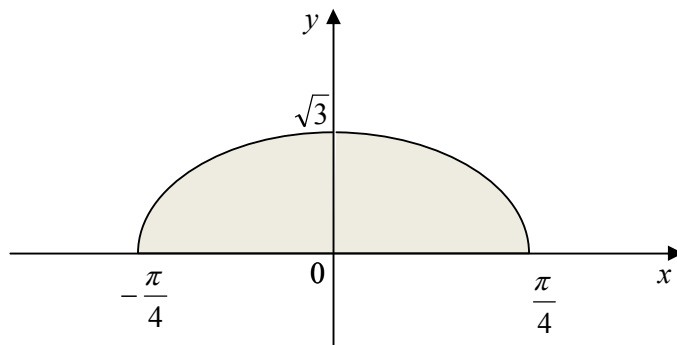
(D) 107

Q9. What are the solutions to the equation $2^{6x} - 9(2^{3x}) + 8 = 0$?

- (A) $x = 1$ or $x = 8$ (B) $x = 0$ or $x = \frac{8}{3}$
 (C) $x = 0$ or $x = 1$ (D) $x = 1$ or $x = \frac{1}{6}$

Q10. The diagram shows the region bounded by the curve $y = \sqrt{3 \cos 2x}$ and the x -axis for $-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$.

The region is rotated about the x -axis to form a solid.



Which of the following gives the volume of the solid?

- (A) $V = 3\pi \int_0^{\frac{\pi}{4}} \cos 2x \, dx$ (B) $V = 9\pi \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \cos 2x \, dx$
 (C) $V = 6\pi \int_0^{\frac{\pi}{4}} \cos 4x \, dx$ (D) $V = 6\pi \int_0^{\frac{\pi}{4}} \cos 2x \, dx$

End of Section I

Section II

Total marks - 90

Attempt Questions 11-16

All questions are of equal value

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

Question 11 (15 marks) Use the Question 11 Writing Booklet **Marks**

(a) Simplify $\frac{2x}{3} - \frac{3x+8}{12}$ /1

(b) Solve $|x-3| \leq 7$ /2

(c) If $h(x) = \begin{cases} ax+1 & \text{if } x \leq 1 \\ x^2-5 & \text{if } x > 1 \end{cases}$ and $h(-2) = h(4)$, find a . /2

(d) If $(3+\sqrt{5})^2 = a + \sqrt{b}$, find a and b . /2

(e) In the diagram below $BC = 1$ unit, $\angle BCA = 60^\circ$ and $\angle CDA = 30^\circ$.

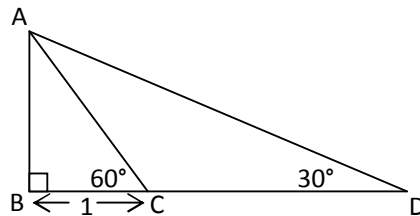


FIGURE NOT
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(i) Find the exact length of AB /1

(ii) Hence, or otherwise, find the length of CD. /2

(f) The first term of a geometric series is 4 and the eighth term is 8748.
Find the twelfth term. /2

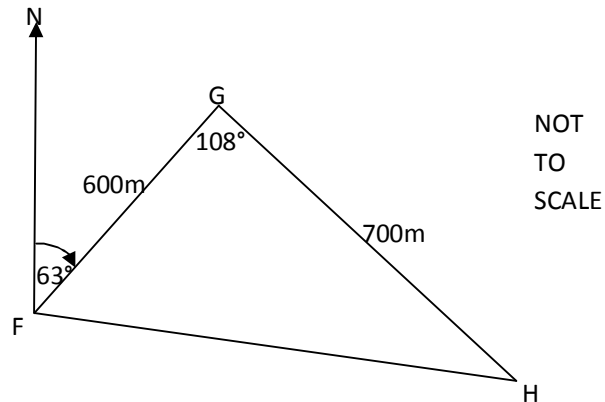
(g) Find the equation of the normal to the curve $y = \tan x$ at the point $(\frac{\pi}{3}, \sqrt{3})$. /3

End of Question 11

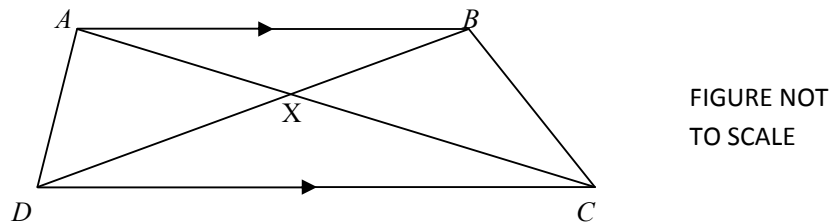
Question 12 (15 marks) Use the Question 12 Writing Booklet.

- (a) Jenny is setting up part of an orienteering course.

She follows the course from F to G to H as shown in the diagram below.



- (i) Show that the distance FH to the nearest metre is 1053 metres. /2
- (ii) Hence, or otherwise, calculate the size of $\angle GFH$ to the nearest degree. /2
- (iii) If the bearing of G from F is 063° calculate the bearing of H from F to the nearest degree. /1
- (b) $ABCD$ is a trapezium, and AB is parallel to DC .
The diagonals AC and BD intersect at X . $AX = 4\text{cm}$, $CX = 10\text{cm}$, $BD = 12\text{cm}$.



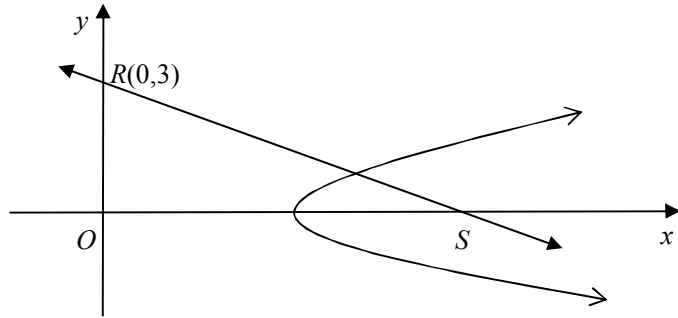
Copy the diagram neatly into your answer booklet.

- (i) Prove $\triangle AXB$ is similar to $\triangle CXD$. /2
- (ii) Find the length of BX . /2

Question 12 continues on page 8

Question 12 continued

- (c) RS is a straight line where R has co-ordinates $(0,3)$ and S is the focus of the parabola $y^2 = 4x - 12$.



- (i) Show that the co-ordinates of the focus S are $(4,0)$. /2
- (ii) Show that the equation of the line RS is $3x + 4y - 12 = 0$ /1
- (iii) Find the perpendicular distance from a point $P(0, \frac{1}{2})$ to the line RS . /2
- (iv) Hence, write the equation of the circle that has its centre at P and has RS as a tangent /1

End of Question 12

Question 13 (15 marks) Use the Question 13 Writing Booklet.

- (a) For the equation $y = \frac{1}{4}x^2 + 2x - 1$
- (i) Show that $(x + 4)^2 = 4(y + 5)$ /2
- (ii) Hence, or otherwise, find:
- (α) the coordinates of the vertex /1
- (β) the equation of the directrix /1
- (b) Show that $\operatorname{cosec}\theta - \sin\theta = \cot\theta \cos\theta$ /2
- (c) Find the volume obtained by rotating about the x axis the area beneath the curve $y = e^x$ from $x = 0$ to $x = 2$. Give your answer as an exact value. /3
- (d) If $1, a, b$ form an arithmetic sequence and $1, b, a$ form a geometric sequence and $a \neq b$:
- (i) Show that $2a - b = 1$ and $a = b^2$ /1
- (ii) Hence, or otherwise, find a and b . /2
- (e) (i) On the same set of axes, sketch the graphs of the functions $y = 2\sin x$ and $y + 1 = 0$ where $-\pi \leq x \leq \pi$. /2
- (ii) Hence, or otherwise, find the number of solutions for $\sin x = -\frac{1}{2}$ in the same domain. /1

End of Question 13

Question 14 (15 marks) Use the Question 14 Writing Booklet.

(a) Find:

(i) $\frac{dy}{dx}$ if $y = x^2 \log_e x$ /2

(ii) $\int \frac{\sin x}{1 + \cos x} dx$ /2

(b) (i) Differentiate $e^{\tan x}$ /1

(ii) Hence, or otherwise, evaluate $\int_0^{\frac{\pi}{4}} e^{\tan x} \sec^2 x \, dx$. /2

(c) Find the value of m for which $x^2 + (m-1)x - m = 0$ has equal roots /3

(d) A rectangular sheet of cardboard measures 12cm by 9cm. From two corners, squares of side x cm are removed as shown. The remainder is folded along the dotted lines to form a tray as shown.

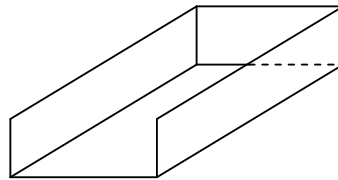
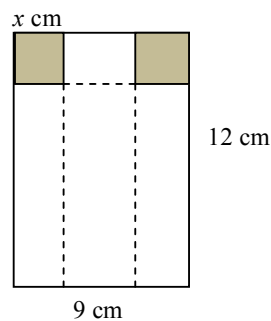


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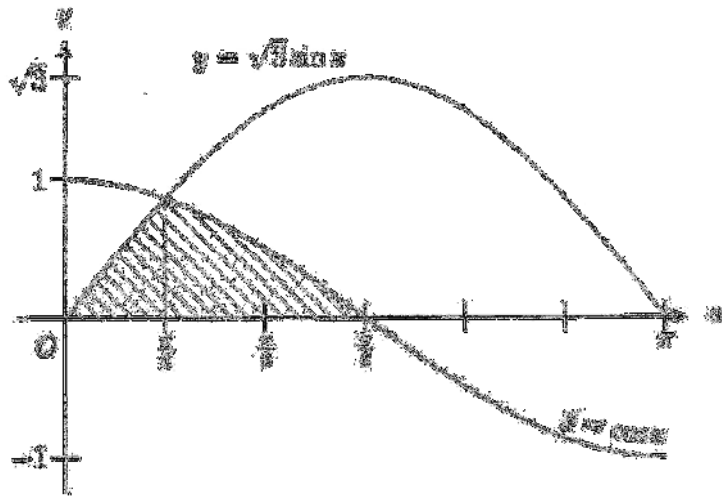
(i) Show that the volume, $V \text{ cm}^3$, of the tray is given by $V = 2x^3 - 33x^2 + 108x$. /2

(ii) Find the maximum volume of the tray /3

End of Question 14

Question 15 (15 marks) Use the Question 15 Writing Booklet.

- (a) The diagram below shows the graphs of the functions $y = \sqrt{3} \sin x$ and $y = \cos x$ between $x = 0$ and $x = \pi$. The two graphs intersect at $x = \frac{\pi}{6}$.



Show that the area of the shaded region between $x = 0$ and $x = \frac{\pi}{6}$ is $(\sqrt{3} - 1)$ square units.

/3

- (b) At the beginning of 2008 the population N of birds on an island was 10 000.

At the beginning of 2012 this population was 160 000.

Assume that the population N was increasing exponentially and satisfies the equation $N = Ae^{kt}$, where A and k are constants and the time t is measured in years from the beginning of 2007.

- (i) Find the constants A and k . /3
- (ii) Find the time t , to the nearest year, required for the population N to reach 1.28 million. /2

Question 15 continues on page 12

Question 15 continued

- (c) The velocity v in ms^{-1} of a particle moving in a straight line is given by:

$$v = 36 - 4e^{2t} \text{ where } t \text{ is the time in seconds and } d \text{ is the distance travelled in metres.}$$

- (i) Find the initial velocity of the particle. /1
- (ii) Show that the exact time at which the particle first comes to rest is $t = \log_e 3$. /2
- (iii) Find the distance travelled by the particle during this time. /2
- (iv) Find an expression for the acceleration a in terms of v . /2

End of Question 15

Question 16 (15 marks) Use the Question 16 Writing Booklet.

- (a) Water is flowing into a container which can be filled to a depth (D) of 900 millimetres. When the water began to flow, the container already held water to a depth of 150 millimetres. The rate, at which the depth of water in the container is increasing, in millimetres per minute, is given by $R = 6t + 5$, ($t \geq 0$). Find:

- (i) the depth of water in the container after eight minutes /3
- (ii) the time it takes to fill the container. /2

- (b) The diagram shows a sector with angle θ at the centre and radius r cm, where $6 < r < 12$. The arc length is 6π cm.

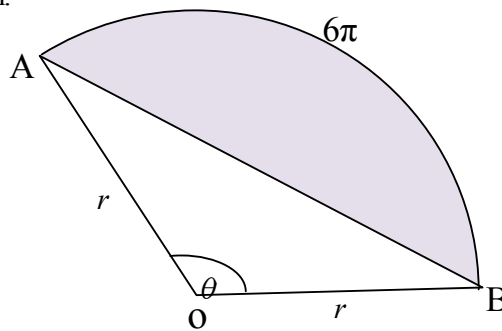


FIGURE NOT
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- (i) Show why θ is an obtuse angle /2
- (ii) Calculate the area of the shaded minor segment when $\theta = \frac{3\pi}{4}$ radians. /2
(Hint: first find the value of r)
- (c) Given the function $y = \frac{10}{3 + 2\sin x}$ in the domain $0 \leq x \leq 2\pi$:
- (i) Show that $\frac{dy}{dx} = \frac{-20\cos x}{(3 + 2\sin x)^2}$ /1
- (ii) Show that there are stationary points at $x = \frac{\pi}{2}$ and $x = \frac{3\pi}{2}$ and describe their nature. /3
- (iii) Graph the function in the given domain showing all essential features /2

End of Examination

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE : $\ln x = \log_e x, \quad x > 0$

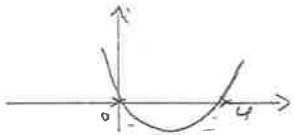
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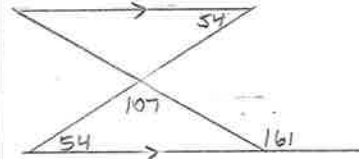
**YEAR 12 TRIAL HSC MATHEMATICS 2014
MULTIPLE CHOICE ANSWER SHEET**

Section I - Mark your answer in the appropriate box with an X.

	A	B	C	D
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Begin Section II using your writing booklets

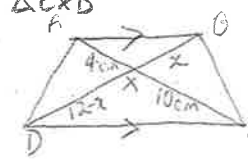
Qn	MC	Solutions	Marks	Comments: Criteria
1.		$x^2y - xy^2 - x + y$ $= xy(x-y) - (x-y)$ $= (x-y)(xy-1)$	B	
2.		$\int (2x+1)^5 dx$ $= \frac{(2x+1)^6}{6 \times 2} + C$ $= \frac{(2x+1)^6}{12} + C$	C	
3.		$1 + \frac{\sqrt{2}}{\sqrt{2}+1} + \frac{2}{(\sqrt{2}+1)^2} + \frac{2\sqrt{2}}{(\sqrt{2}+1)^3} + \dots$ $a=1 \quad r = \frac{\sqrt{2}}{\sqrt{2}+1}$ $S_{\infty} = \frac{a}{1-r}$ $= \frac{1}{1 - \left(\frac{\sqrt{2}}{\sqrt{2}+1}\right)}$ $= \frac{1}{\frac{(\sqrt{2}+1) - \sqrt{2}}{\sqrt{2}+1}}$ $= \sqrt{2}+1$	A	
4.		$kx^2 + kx + 1 = 0$ $a=k \quad b=k \quad c=1$ $b^2 - 4ac < 0$ $k^2 - 4 \times k \times 1 < 0$ $k^2 - 4k < 0$ $k(k-4) < 0$ $\therefore 0 < k < 4$	B	

Qn	Solutions	Marks	Comments: Criteria
5.	$\sqrt{3} \tan x + 3 = 0$ $\sqrt{3} \tan x = -3$ $\tan x = \frac{-3}{\sqrt{3}}$ $= \frac{-3\sqrt{3}}{3}$ $= -\sqrt{3}$ $\text{acute angle} = \frac{\pi}{3}$ $\therefore \theta = \pi - \frac{\pi}{3}, 2\pi - \frac{\pi}{3}$ $= \frac{2\pi}{3}, \frac{5\pi}{3}$	C	
6.	$A \neq \frac{0.5}{3} \{4 + 4(1+3) + 2(-2) + 8\}$ $= \frac{1}{6} \{4 + 16 - 4 + 8\}$ $= \frac{1}{6} \{24\}$ $= 4$	D	
7.	$2x^2 + 3x - 6 = 0$ $a=2 \quad b=3 \quad c=-6$ $\alpha + \beta = \frac{-b}{a} \quad \alpha\beta = \frac{c}{a}$ $= \frac{-3}{2} \quad = \frac{-6}{2}$ $= -\frac{3}{2} \quad = -3$ $\therefore \frac{\alpha\beta}{\alpha+\beta} = \frac{-3}{-\frac{3}{2}}$ $= -3 \times \frac{2}{-3}$ $= 2$	A	
8.	 $\theta = 161^\circ$	A	

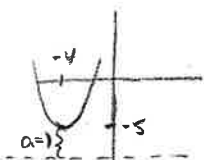
Qn	Solutions	Marks	Comments: Criteria
9.	$2^{6x} - 9(2^{3x}) + 8 = 0$ ie $(2^{3x})^2 - 9(2^{3x}) + 8 = 0$ let $u = 2^{3x}$ $u^2 - 9u + 8 = 0$ $(u-8)(u-1) = 0$ $2^{3x} - 8 = 0$ or $2^{3x} - 1 = 0$ $2^{3x} = 8$ $2^{3x} = 1$ $2^{3x} = 2^3$ $2^{3x} = 2^0$ $3x = 3$ $3x = 0$ $x = 1$ $x = 0$		C
10.	$V = \pi \int_a^b y^2 dx$ $= \pi \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} (3 \cos 2x)^2 dx$ $= \pi \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} 9 \cos^2 2x dx$ $= 9\pi \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \cos^2 2x dx$ $= 9\pi \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \frac{1 + \cos 4x}{2} dx$ $= \frac{9\pi}{2} \left[x + \frac{\sin 4x}{4} \right]_{-\frac{\pi}{4}}^{\frac{\pi}{4}}$ $= \frac{9\pi}{2} \left[\frac{\pi}{4} + \frac{\sin \pi}{4} - \left(-\frac{\pi}{4} + \frac{\sin -\pi}{4} \right) \right]$ $= \frac{9\pi}{2} \left[\frac{\pi}{4} + 0 - \left(-\frac{\pi}{4} + 0 \right) \right]$ $= \frac{9\pi}{2} \left[\frac{\pi}{4} + \frac{\pi}{4} \right]$ $= \frac{9\pi}{2} \left[\frac{2\pi}{4} \right]$ $= \frac{9\pi}{2} \left[\frac{\pi}{2} \right]$ $= \frac{9\pi^2}{4}$		D

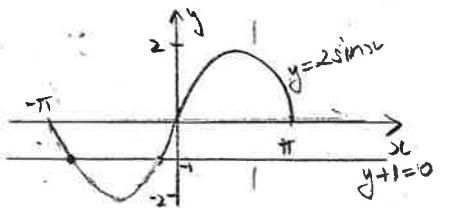
Qn	Solutions	Marks	Comments: Criteria
(a)	$\frac{2x}{3} - \frac{3x+8}{12} = \frac{8x-3x-8}{12}$ $\frac{2x}{3} - \frac{3x+8}{12} = \frac{5x-8}{12}$	1	$\frac{1}{2}$ For $8x-3x+8$ $= \frac{5x+8}{12}$
(b)	$ x-3 \leq 7$ $-7 \leq x-3 \leq 7$ $-4 \leq x \leq 10$	2	$x-3 \leq 7$ or $-(x-3) \leq 7$ $x \leq 10$, $-x+3 \leq 7$ $-x \leq 4$ $-4 \leq x \leq 10$
(c)	$h(-2) = -2a+1$ $h(4) = 4^2-5 = 11$ If $h(-2) = h(4)$ $-2a+1 = 11$ $-2a = 10$ $a = -5$	2	
(d)	$(3+\sqrt{5})^2 = 9 + 6\sqrt{5} + 5$ $= 14 + 6\sqrt{5}$ $= 14 + \sqrt{180}$	2	$\therefore a = \frac{1}{2} \quad b = \frac{1}{2}$
(e)	i) $\tan 60^\circ = \frac{AB}{1}$ $\therefore AB = \sqrt{3}$ ii) $\tan 30^\circ = \frac{AB}{BD}$ $BD = \frac{\sqrt{3}}{\tan 30^\circ}$ $= \frac{\sqrt{3}}{\frac{1}{\sqrt{3}}}$ $= 3$ $\therefore CD = BD - BC$ $= 3 - 1$ $= 2 \text{ units}$	2	
(f)	$a = 4$ $ar^7 = 8748$ $4r^7 = 8748$ $r^7 = 2187$ $r = 3$ $\therefore T_{12} = ar^{11}$ $= 4 \times 3^{11}$ $= 708588$	2	$\frac{1}{2}$ IF DONE AS AN A.P.

Qn	Solutions	Marks	Comments: Criteria
11 cont.			
(9)	$y = \tan x$ $\frac{dy}{dx} = \sec^2 x$ at $(\frac{\pi}{3}, \sqrt{3})$ $\frac{dy}{dx} = \sec^2(\frac{\pi}{3})$ $= (2)^2 = 4$ $\therefore m_{\perp} = -\frac{1}{4}$ $y - y_1 = m(x - x_1)$ $y - \sqrt{3} = -\frac{1}{4}(x - \frac{\pi}{3})$ $4y - 4\sqrt{3} = -x + \frac{\pi}{3}$ $x + 4y - 4\sqrt{3} - \frac{\pi}{3} = 0$ $3x + 12y - 12\sqrt{3} - \pi = 0$ $(\text{or } y = -\frac{x}{4} + \sqrt{3} + \frac{\pi}{12})$	3	

Qn	Solutions	Marks	Comments: Criteria
12			
(i)	$FH^2 = 600^2 + 700^2 - 2 \times 600 \times 700 \cos 108^\circ$ $= 1109574.275$ $FH = 1053$ (nearest metre)	2	
(ii)	$\frac{\sin \theta}{700} = \frac{\sin 108^\circ}{1053}$ $\sin \theta = \frac{\sin 108^\circ}{1053} \times 700$ $\therefore \theta = 39^\circ$	2	
(iii)	bearing of H from F is $63 + 39 = 102^\circ$	1	
(b)	In $\triangle AXB$ and $\triangle CXD$		
(i)	$\angle BAX = \angle DCX$ (alternate $\angle$, $AB \parallel DC$) $\angle ABX = \angle CDX$ (alternate $\angle$, $AB \parallel DC$) $\angle AXB = \angle CXD$ (vert. opp) $\therefore \triangle AXB \parallel \triangle CXD$ (equiangular)	2	
(ii)	Since $\triangle AXB \parallel \triangle CXD$  $\frac{BX}{DX} = \frac{AX}{CX}$ $\frac{4x}{10-x} = \frac{x}{12-x}$ $14x = 48$ $BX = \frac{24}{7}$	2	For $\frac{BX}{DX} = \frac{4}{10}$
(c)	$y^2 = 4x - 12$ $y^2 = 4(x - 3)$ Vertex is $(3, 0)$ $4a = 4$ $a = 1$ $\therefore$ focus is $(4, 0)$	2	
(ii)	$m_{RS} = -\frac{3}{4}$ $y - 0 = -\frac{3}{4}(x - 4)$ $4y = -3x + 12$ $3x + 4y - 12 = 0$	1	

Qn	Solutions	Marks	Comments: Criteria
12			
c)			
iii)	$d = \frac{ ax_1 + by_1 + c }{\sqrt{a^2 + b^2}}$ $= \frac{ 3(0) + 4(\frac{1}{2}) - 12 }{\sqrt{3^2 + 4^2}}$ $= \frac{ -10 }{\sqrt{25}}$ $= 2$	2	
iv)	$\text{circle: } (x-0)^2 + (y-\frac{1}{2})^2 = 4$	1	1/2 FOR WRONG RADIUS

Qn	Solutions	Marks	Comments: Criteria
13			
(a)			
i)	$y = \frac{1}{4}x^2 + 2x - 1$ $4y = x^2 + 8x - 4$ $4y + 4 = x^2 + 8x \quad \left\{ \frac{1}{2} \right.$ $4y + 4 + \left(\frac{8}{2}\right)^2 = x^2 + 8x + \left(\frac{8}{2}\right)^2$ $4y + 20 = (x+4)^2$ $\therefore (x+4)^2 = 4(y+5)$	2	
ii)	$A) \text{ vertex} = (-4, -5)$ $B) \quad 4a = 4$ $a = 1 \quad (\text{focal length})$ $\therefore \text{directrix} = y = -6$ 	1	
(b)	$\operatorname{cosec} \theta - \sin \theta = \frac{1}{\sin \theta} - \sin \theta$ $= \frac{1 - \sin^2 \theta}{\sin \theta}$ $= \frac{\cos^2 \theta}{\sin \theta} \quad \frac{1}{2}$ $= \frac{\cos \theta}{\sin \theta} \times \cos \theta \quad \frac{1}{2}$ $= \cot \theta \cos \theta$	2	
(c)	$A = \pi \int_0^2 y^2 dx$ $= \pi \int_0^2 (e^x)^2 dx \quad \frac{1}{2}$ $= \pi \int_0^2 e^{2x} dx \quad \frac{1}{2}$ $= \pi \left[\frac{e^{2x}}{2} \right]_0^2$ $= \pi \left(\frac{e^4}{2} - \frac{e^0}{2} \right)$ $= \pi \left(\frac{e^4}{2} - \frac{1}{2} \right)$	3	

Qn	13	Solutions	Marks	Comments: Criteria
(d)		$a-1=b-a$ ① $\therefore 2a-b=1$ $\frac{b}{1}=\frac{a}{b}$ ② $b^2=a$	1	
(ii)		sub ② into ① $2(b^2)-1=b$ ③ $2b^2-b-1=0$ $(2b+1)(b-1)=0$ ④ $b=-\frac{1}{2}, 1$ $\therefore b=-\frac{1}{2}$ and $a=\frac{1}{4}$	2	$-\frac{1}{2}$ if both solutions are given.
(e)				
(i)			2	
(ii)		$2 \sin x = -1$ $\sin x = -\frac{1}{2}$	1	$\therefore$ 2 solutions

Qn	14	Solutions	Marks	Comments: Criteria
(a)		$y = x^2 \log_e x$ $u = x^2$ $v = \log_e x$ $\frac{du}{dx} = 2x$ $\frac{dv}{dx} = \frac{1}{x}$		
(i)		$\frac{dy}{dx} = 2x \log_e x + x^2 \times \frac{1}{x}$ $= 2x \log_e x + x$ $= x(2 \log_e x + 1)$	2	
(ii)		$\int \frac{\sin x}{1 + \cos x} dx = -\int \frac{-\sin x}{1 + \cos x} dx$ $= -\log_e (1 + \cos x) + c$	2	1 for log 1 for minus
(b)				
(i)		$\frac{d}{dx} (e^{\tan x}) = \sec^2 x \tan x$	1	
(ii)		$\int_0^{\frac{\pi}{4}} \frac{d}{dx} e^{\tan x} dx = \int_0^{\frac{\pi}{4}} \sec^2 x \tan x dx$ $e^{\tan x} \Big _0^{\frac{\pi}{4}} = \int_0^{\frac{\pi}{4}} \sec^2 x \tan x dx$ $e^{\tan \frac{\pi}{4}} - e^{\tan 0} = \frac{1}{2}$ $e^1 - e^0 =$ $\therefore \int_0^{\frac{\pi}{4}} \sec^2 x \tan x dx = e - 1$	2	1 for $\int_0^{\frac{\pi}{4}} e^{\tan x} dx$ $\frac{1}{2}$ for sub. $\frac{1}{2}$ for answer
(c)		$x^2 + (m-1)x - m = 0$ $x + x = -\frac{b}{a}$ $x \times x = \frac{c}{a}$ $2x = -\frac{(m-1)}{1}$ $x^2 = -\frac{m}{1}$ ② $2x = 1 - m$ $x = \frac{1-m}{2}$ ① sub ① into ② $\left(\frac{1-m}{2}\right)^2 = -m$ $\frac{1-2m+m^2}{4} = -m$ $1-2m+m^2 = -4m$ $m^2+2m+1=0$ $(m+1)^2=0$ $\therefore m=-1$		
		OR $b^2 - 4ac = 0$ $(m-1)^2 - 4 \times 1 \times (-m) = 0$ $m^2 - 2m + 1 + 4m = 0$ $m^2 + 2m + 1 = 0$ $(m+1)^2 = 0$ $m = -1$	3	

Qn	15 Cont	Solutions	Marks	Comments: Criteria
(c)				
(i)	$v = 36 - 4e^{2t}$ $t=0, v = 36 - 4e^{2(0)} \quad \checkmark$ $v = 36 - 4(1)$ $= 32 \text{ m/s}$	1		
(ii)	When $v=0$ $36 - 4e^{2t} = 0 \quad \checkmark \quad \frac{1}{2}$ $36 = 4e^{2t} \quad \frac{1}{2}$ $9 = e^{2t}$ $\log_e 9 = \log_e e^{2t} \quad \frac{1}{2}$ $\log_e 9 = 2t \log_e e$ $\frac{\log_e 9}{2} = t$ $\frac{2 \log_e 3}{2} = t \quad \frac{1}{2}$ $\therefore t = \log_e 3$	2	$\frac{1}{2}$ for $=0$	
(iii)	$d = \int v \, dt$ $= \int_0^{\log_e 3} (36 - 4e^{2t}) \, dt \quad \checkmark \quad \frac{1}{2}$ $= \left[36t - \frac{4e^{2t}}{2} \right]_0^{\log_e 3} \quad \checkmark \quad \frac{1}{2}$ $= \left(36(\log_e 3) - \frac{4e^{2 \log_e 3}}{2} \right) - \left(36(0) - \frac{4e^{2(0)}}{2} \right) \quad \frac{1}{2}$ $= (36 \log_e 3 - 2e^{\log_e 9}) - (0 - 2)$ $= 36 \log_e 3 - 2 \times 9 + 2$ $= (36 \log_e 3 - 16) \text{ m} \quad \frac{1}{2}$ $= 23.55 \text{ m}$	2		
(iv)	$a = \frac{dv}{dt}$ $= -8e^{2t} \quad \textcircled{1} \quad \checkmark$ $v = 36 - 4e^{2t}$ $4e^{2t} = 36 - v \quad \frac{1}{2}$ $e^{2t} = \frac{36-v}{4} \quad \textcircled{2}$ sub $\textcircled{2}$ into $\textcircled{1}$ $a = -8 \left(\frac{36-v}{4} \right) \quad \frac{1}{2}$ $= -2(36-v)$	2		

or 15 c iii)

$$v = 36 - 4e^{2t}$$

$$d = \int (36 - 4e^{2t}) \, dt \quad \frac{1}{2}$$

$$d = 36t - 2e^{2t} + c$$

at $t=0, d=0$

$$0 = 36(0) - 2e^{2(0)} + c$$

$$c = 2 \quad \frac{1}{2}$$

at $t = \log_e 3$

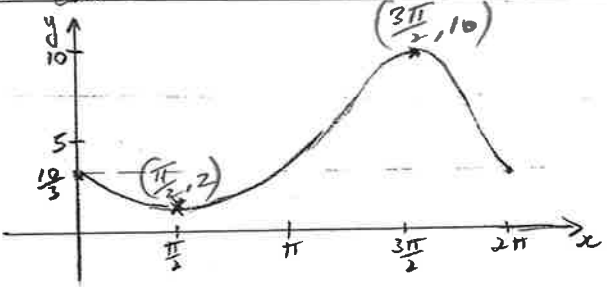
$$d = 36(\log_e 3) - 2e^{2 \log_e 3} + 2 \quad \frac{1}{2}$$

$$= 36(\log_e 3) - 2 \times 9 + 2$$

$$= 36 \log_e 3 - 16 \quad \frac{1}{2}$$

Qn	16	Solutions	Marks	Comments: Criteria
(a)				
(i)		$R = 6t + 5$ $D = \int R \, dt$ $= \int 6t + 5 \, dt$ $\therefore D = \frac{6t^2}{2} + 5t + C$ when $t=0$ $D=150$ $150 = 3(0)^2 + 5(0) + C$ $\therefore C = 150$ $\therefore D = 3t^2 + 5t + 150$ when $t=8$ $D = 3(8)^2 + 5(8) + 150$ $= 382 \text{ m}$	3	
(ii)		when $D=900$ $900 = 3t^2 + 5t + 150$ $3t^2 + 5t + 750 = 0$ $(3t+50)(t-15) = 0$ $\therefore t = 15 \text{ since } t > 0$	2	$\frac{1}{2}$ if give $t = -\frac{50}{3}$ as an answer.
(b)				
(i)		$6\pi = r\theta$ $\frac{6\pi}{r} = \theta$ since $6 < r < 12$ if $r=6$ $\theta = \frac{6\pi}{6} = \pi$ if $r=12$ $\theta = \frac{6\pi}{12} = \frac{\pi}{2}$ $\therefore \frac{\pi}{2} < \theta < \pi$ so θ is obtuse	2	1 for $r=8$ $\frac{1}{2}$ for substit.
(ii)		if $\theta = \frac{3\pi}{4}$ $6\pi = \frac{3\pi}{4} r$ $\therefore r = 8$ $A = \frac{1}{2} r^2 (\theta - \sin \theta)$ $= \frac{1}{2} (8)^2 \left(\frac{3\pi}{4} - \sin \frac{3\pi}{4} \right)$ $= 32 \left(\frac{3\pi}{4} - \frac{1}{\sqrt{2}} \right)$ $= 24\pi - 16\sqrt{2}$ $= 52.77$	2	$\frac{1}{2}$ for correct simplified answer

Qn	16 cont	Solutions	Marks	Comments: Criteria
(c)				
(i)		$y = \frac{10}{3+2\sin x} \quad 0 \leq x \leq 2\pi$ $y = 10(3+2\sin x)^{-1}$ $\frac{dy}{dx} = -10 \cos x (3+2\sin x)^{-2} \times -2$ $= \frac{-20 \cos x}{(3+2\sin x)^2}$	1	
(ii)		for stationary pts $\frac{dy}{dx} = 0$ $\frac{-20 \cos x}{(3+2\sin x)^2} = 0$ $-20 \cos x = 0$ $\cos x = 0$ $\therefore x = \frac{\pi}{2}, \frac{3\pi}{2}$ $x = \frac{\pi}{2}$ $y = \frac{10}{3+2\sin \frac{\pi}{2}} = \frac{10}{3+2} = \frac{10}{5} = 2$ $\left(\frac{\pi}{2}, 2\right)$ is a minimum $x = \frac{3\pi}{2}$ $y = \frac{10}{3+2\sin \frac{3\pi}{2}} = \frac{10}{3+2(-1)} = \frac{10}{3+2(-1)} = \frac{10}{1} = 10$ $\left(\frac{3\pi}{2}, 10\right)$ is a maximum	3	if only points $\frac{1}{2}$ $\frac{1}{2}$ for nature only $\frac{2\frac{1}{2}}{3}$ if don't find points $2\frac{1}{2}$ for wrong min or max
(iii)		at $x=0$ $y = \frac{10}{3+2\sin 0} = \frac{10}{3}$ at $x=2\pi$ $y = \frac{10}{3+2\sin 2\pi} = \frac{10}{3}$		

Qn	16 cont.	Solutions	Marks	Comments: Criteria
			2	$-\frac{1}{2}$ for wrong or no endpoints. -1 for wrong shape $-\frac{1}{2}$ for wrong or no scale